

Data Analytics Interview Question Bank

SQL • PYTHON/R • STATISTICS • DATA MODELING

Target Role: Data Analyst / Analytics Engineer

Total Questions: 80 (10 per Level per Domain)

Domain 1: SQL Data Manipulation & Database Analytics

LEVEL 1: FOUNDATIONAL

1. What is the execution order of a standard SQL query containing SELECT, FROM, WHERE, GROUP BY, HAVING, and ORDER BY?
2. What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN?
3. How does the WHERE clause differ from the HAVING clause when filtering aggregations?
4. What are NULL values in SQL, and how do functions like COALESCE() and IS NULL handle them?
5. What is the difference between UNION and UNION ALL in terms of performance and output?
6. What is the primary key versus a foreign key, and why are referential integrity constraints important?
7. How do aggregate functions like COUNT(), SUM(), AVG(), MIN(), and MAX() treat NULL values?
8. What is a subquery, and what is the difference between a scalar subquery and a correlated subquery?
9. How do string manipulation functions like CONCAT(), SUBSTRING(), and LIKE pattern matching work in SQL?
10. What is the purpose of the DISTINCT keyword, and how does it affect query execution time?

LEVEL 2: INTERMEDIATE / PROJECT & RESUME BASED

11. How do window functions like ROW_NUMBER(), RANK(), and DENSE_RANK() differ when ranking tied values?
12. How do you compute a 7-day rolling average of daily user acquisitions using SQL window frames (ROWS BETWEEN)?
13. Describe how Common Table Expressions (CTEs) improve query readability and optimization over nested subqueries.
14. How would you write a query to detect duplicate records across multiple columns without altering the source table?
15. How do you optimize an analytical SQL query that is running slowly on a table with 50+ million records?
16. How do LEAD() and LAG() window functions allow you to perform year-over-year or month-over-month growth calculations?
17. Explain how conditional aggregation using CASE WHEN inside SUM() or COUNT() can pivot row data into columns.
18. How do you identify sessionization gaps (user inactivity periods exceeding 30 minutes) using SQL windowing logic?
19. What is the difference between clustered and non-clustered indexes, and how do they impact read vs. write performance?
20. How do you handle date formatting, timezone conversions, and interval arithmetic across cross-geography data?

LEVEL 3: ADVANCED / PRODUCTION & SCENARIO BASED

21. Write a query framework to perform a cohort retention analysis tracking month-over-month user activity for 12 months post-signup.
22. How do you handle severe data skew in distributed SQL engines (e.g., Snowflake, BigQuery, Spark SQL) during heavy JOIN operations?
23. How would you design a query to calculate customer lifetime value (LTV) across non-contiguous subscription renewal cycles?

24. Explain how indexing strategies (B-Tree, Hash, Bitmap) impact query performance on Read-Heavy OLAP vs. Write-Heavy OLTP systems.
25. How do you debug data discrepancies caused by fan-out issues during multi-table LEFT JOINS containing One-to-Many relationships?
26. How do you write recursive CTEs to traverse hierarchical organizational structures or parent-child product categories?
27. How do partition pruning and clustering keys optimize query costs and speed in cloud data warehouses like BigQuery?
28. How do you handle slowly changing dimensions (SCD Type 1 vs. Type 2 vs. Type 3) when tracking historical customer updates?
29. How do materialized views differ from standard views, and how do you set up automated refresh schedules in production?
30. How do you analyze semi-structured JSON payloads directly inside SQL using JSON_EXTRACT or FLATTEN array operations?

LEVEL 4: EXPERT / HIGH-PRESSURE & STRATEGY SCENARIOS

31. A critical C-suite dashboard suddenly shows a 30% drop in revenue metrics post-migration. How do you triage the pipeline lineage and SQL transformation logic in 2 hours?
32. How do you architect an enterprise Data Mart schema (Star Schema vs. Snowflake Schema) to support thousands of concurrent ad-hoc analytics queries?
33. How do you enforce automated data quality testing checks (e.g., Great Expectations, dbt tests) across multi-stage SQL transformations before data reaches BI tools?
34. Your organization's Snowflake/BigQuery cloud bill increased by 300% this month. What is your strategy for auditing query costs and enforcing governance?
35. How do you design a zero-downtime database migration schema strategy for a production enterprise platform with continuous active writes?
36. How do you handle near-real-time streaming analytics query models vs. traditional batch processing data models?
37. How do you design data security models including row-level security (RLS) and dynamic column masking for PII compliance?
38. How do you resolve transactional deadlock cascades in distributed multi-region databases supporting high write volumes?
39. How do you build automated query performance monitoring dashboards to proactively identify degrading ETL jobs?
40. What is your framework for establishing company-wide SQL coding standards, dbt documentation, and data dictionary governance?

Domain 2: Python / R for Data Modeling & Statistics

LEVEL 1: FOUNDATIONAL

41. What is the difference between pandas Series and DataFrames, and how are vectorised operations executed?
42. What is the difference between Mean, Median, and Mode, and when is Median preferred over Mean?
43. How do standard deviation and variance measure data dispersion in a distribution?
44. What is the Central Limit Theorem, and why is it essential for inferential statistics?
45. How do you handle missing values in a dataset using imputation techniques vs. deletion?
46. What is the difference between loc[] and iloc[] indexing methods in pandas?

- 47. What is the difference between population parameter and sample statistic?
- 48. How do Pearson correlation and Spearman rank correlation differ?
- 49. What are Probability Density Functions (PDF) vs. Cumulative Distribution Functions (CDF)?
- 50. How do apply(), map(), and applymap() functions differ in pandas?

LEVEL 2: INTERMEDIATE / PROJECT & RESUME BASED

- 51. Explain p-value, Type I error, and Type II error in the context of statistical hypothesis testing.
- 52. How do you identify and treat outliers in a skewed distribution using the IQR method and Z-score thresholding?
- 53. Walk through a project where you used Python (Pandas/NumPy) to clean messy unstructured JSON event logs into structured analytical tables.
- 54. What is multi-collinearity in Linear Regression, and how do you detect it using Variance Inflation Factor (VIF)?
- 55. How do you evaluate a classification model using Precision, Recall, F1-Score, and ROC-AUC curves?
- 56. How do feature scaling techniques like Min-Max Scaling and Standardization (Z-score) impact statistical algorithms?
- 57. Explain One-Hot Encoding versus Ordinal Encoding, and when each approach is appropriate.
- 58. How do you use Chi-Square tests of independence to evaluate relationships between categorical variables?
- 59. How do memory optimization techniques (e.g., downcasting numeric types, categorical categories) prevent memory errors in pandas?
- 60. How do you perform K-fold cross-validation to prevent model overfitting on small sample sizes?

LEVEL 3: ADVANCED / PRODUCTION & SCENARIO BASED

- 61. How would you set up an A/B test analysis script that accounts for sample size calculation, minimum detectable effect (MDE), and Bonferroni corrections?
- 62. How do you model customer churn prediction when dealing with extreme class imbalance (95% non-churn vs. 5% churn)?
- 63. How do you decompose time-series data into trend, seasonality, and residual components for demand forecasting?
- 64. Explain the difference between parametric and non-parametric statistical tests (e.g., T-Test vs. Mann-Whitney U Test) and when to apply each.
- 65. How do you apply Principal Component Analysis (PCA) for dimensionality reduction while preserving maximum explained variance?
- 66. How do you optimize hyperparameter tuning algorithms (Grid Search vs. Random Search vs. Bayesian Optimization) for large models?
- 67. How do you evaluate survival analysis models (e.g., Kaplan-Meier curves, Cox Proportional Hazards) for user retention modeling?
- 68. How do you set up bootstrap resampling methods to estimate confidence intervals for complex non-standard metrics?
- 69. How do you parallelize large-scale pandas operations using Dask, Ray, or PySpark when data size exceeds system RAM?
- 70. How do you construct Markov Chain attribution models for evaluating multi-touch digital marketing campaigns?

LEVEL 4: EXPERT / HIGH-PRESSURE & STRATEGY SCENARIOS

- 71. Executive stakeholders reject a machine learning churn model because it acts as a "black box." How do you use SHAP/LIME values to explain feature influence?
- 72. How do you establish automated data drift and concept drift monitoring for machine learning models running in production analytical pipelines?

73. How do you design a causal inference analytics framework (e.g., Difference-in-Differences, Synthetic Control) when randomized A/B testing is impossible?
74. An A/B test returns statistically significant positive results, but overall business revenue drops post-rollout. How do you diagnose Simpson's Paradox or novelty effects?
75. How do you architect an enterprise automated anomaly detection system for financial fraud monitoring in real-time streaming data?
76. How do you design uplift modeling algorithms to target incremental customers who will convert only if exposed to an incentive?
77. How do you evaluate Bayesian statistical modeling against Frequentist approaches for executive decision-making under uncertainty?
78. How do you manage data governance, reproducible research environments, and model registries across distributed analytics teams?
79. How do you build automated predictive forecasting pipelines that gracefully handle extreme macro-economic disruptions or black swan events?
80. How do you translate high-dimensional cluster analysis results into actionable business persona strategies for executive leadership?